A Dual-Signal Early-Warning System for Indian Bank and NBFC Financial Distress

Combining Accounting-Based Distress Scoring with Merton Structural Credit Risk

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563 institutions - 6,223 company-years - FY2006-FY2023

303,995 daily closing prices - 18 labelled distress events

All inputs public; full pipeline reproducible

Academic decision support. Not investment advice and not a credit rating.

Outline

- **I. Introduction** — motivation and problem statement
- **II. Background** — structural and accounting approaches to default risk
- **III. Data** — sources, construction, and recovery of missing institutions
- **IV. Methodology** — the distress index, the Merton model, and the alert rule
- **V. Results** — lead times, discrimination, and the alert-utility test
- **VI. External validation** — the frozen method on 547 US bank failures
- **VII. Discussion** — signals that failed, and why
- **VIII. Limitations**
- **IX. Conclusion**

Failures are announced on one day; they develop over years

- India's financial sector saw repeated sudden distress: IL&FS (2018), DHFL (2019), Yes Bank (2020), Reliance Capital (2021), SREI (2021).
- Credit ratings and regulatory action largely *confirm* distress rather than anticipate it; commercial early-warning tools are proprietary.
- Yet most of the deterioration is visible in filings and prices that are already public.

> **Research question.** Using only public accounting statements and market prices, can institution-level distress be detected early enough, and selectively enough, to be operationally useful?

Two families of default models, with opposite blind spots

	Accounting-based	Structural (market-based)
Origin	Altman Z-score [2]	Merton [1]; Black-Scholes [5]
Input	Financial statements	Equity value and volatility, debt
Mechanism	Ratios of capital, leverage, profitability	Equity as a call option on firm assets
Coverage	Any institution that files	Listed firms with a liquid quote
Blind spot	Inherits whatever filings conceal	Needs a market; ignores state support

TABLE I. Complementary properties of the two model families.

- Bharath and Shumway [6] show the Merton distance-to-default functional form carries most of its predictive content — motivating its use as an independent second signal.
- This work runs both, and treats their *disagreement* as the informative quantity.

Three public sources, combined point-in-time

Source	Content	Scale
NSE/BSE company financials archive	Balance sheet, P&L, cash flow, ratios	556 institutions
Recovered from source filings	Institutions the archive dropped after merger or resolution	7 institutions
NSE daily bhavcopy	Daily closing prices and corporate actions	303,995 observations, 180 symbols
RBI releases and public filings	Dated distress events	18 events

TABLE II. Data sources.

- **Survivorship bias identified and partly corrected.** Of 22 known distress episodes, only 11 were evaluable; institutions that merge or resolve leave the archive. Seven were recovered from source filings in identical accounting definitions, raising coverage to **18 of 22 (82%)**.
- **Point-in-time discipline.** Accounts for a fiscal year ending 31 March are treated as public from 30 June, so no result uses information unavailable at the time of prediction.

What the raw records actually look like

Each institution is a folder of eight CSV files — balance sheet, profit and loss, cash flow and ratios, yearly and quarterly. Figures are in Rs crore. Extract, *Yes Bank Ltd / Yearly_Balance_Sheet.csv*:

Line item	FY2017	FY2018	FY2019	FY2020
Equity Capital	456	461	463	2,510
Reserves	21,583	25,292	26,424	19,185
Borrowings	181,464	275,582	335,982	219,102
Total Assets	215,060	312,450	380,860	257,832

TABLE III. Source records, Yes Bank Ltd.

The same records for a bank that did fail. Reserves more than halve and the funding base contracts 40% in two years — deposit flight visible in the raw filing, before any modelling:

Line item	FY2017	FY2018	FY2019	FY2020
Reserves	1,945	2,072	**1,573**	**893**
Borrowings	32,326	37,322	30,201	**22,199**
Total Assets	35,245	40,429	33,046	**24,422**

TABLE IV. Lakshmi Vilas Bank Ltd. (PCA Sep 2019, amalgamated Nov 2020), recovered from source filings.

Fundamental Distress Index (FDI)

Six ratios, each converted to a within-year percentile against comparable institutions, then weighted:

Component	Definition	Weight
Capital buffer	equity / total assets	0.30
Capital erosion	$(\text{equity}_t - \text{equity}_{t-2}) / \text{assets}_t$	0.20
Profitability	net profit / total assets	0.20
Leverage	borrowings / equity	0.12
Interest burden	interest expense / revenue	0.10
Earnings trend	$\text{ROA}_t - \text{ROA}_{t-3}$	0.08

TABLE V. Index components.

Peer-relative

Banks are ranked against banks, NBFCs against NBFCs of similar size. Ranking all institutions together penalised every normally funded lender.

Dual-anchored

Capital components are additionally ranked against the whole sector, so a uniformly thin group does not look safe by its own standards.

Not fitted

Weights come from economic reasoning. With 18 events, fitted weights would memorise the sample.

From filing to score: Yes Bank, FY2019

Accounts public 30 June 2019; RBI moratorium 5 March 2020. Ranked against the other 38 banks.

```
equity   =     463 + 26,424  =   26,887      ROA      =   1,709 / 380,860 = 0.0045
assets   =                         380,860      leverage = 335,982 / 26,887  = 12.50
erosion  = (26,887 - 22,039) / 380,860 = 0.0127      [FY2017 equity 22,039]
```

Component	Raw value	Percentile	Weight	Contribution
Capital buffer	0.0706	78.7	0.30	23.62
Capital erosion	0.0127	57.1	0.20	11.43
Profitability (ROA)	0.0045	51.3	0.20	10.26
Leverage	12.50	80.4	0.12	9.65
Interest burden	0.6687	71.8	0.10	7.18
Earnings trend	-0.0108	85.3	0.08	6.82
Distress index				**68.96**

TABLE VI. Component percentiles and weighted contributions.

69 is below the level threshold of 80, so a level-only screen stays silent. But Yes Bank's own three-year median was **39.9**, making this a **+29.0** move against itself — the deterioration leg fires **8.2 months before the moratorium**. Profitability still ranked 51st percentile: the accounts did not yet show the problem.

Merton structural credit risk

Equity is a call option on firm assets struck at the face value of debt. Two equations are solved simultaneously for the two unobservable quantities, asset value V and asset volatility σ_V:

```
E  =  V N(d1)  -  D e^(-rT) N(d2)
sE =  (V/E) N(d1) sV

DD = [ln(V/D) + (r - sV^2/2)T] / (sV sqrt(T))      PD = N(-DD)
```

- Solved by fixed-point iteration (the KMV approach [3]); non-convergence returns no estimate rather than a partial one.
- Market capitalisation uses raw prices with share count from equity capital and split-adjusted face value; volatility uses fully adjusted returns over 250 days.
- **Corporate actions were the dominant engineering problem.** Missed stock splits produced 170% spurious volatility. Detection uses single-day price ratios that land on exact fractions, with guards for circuit limits, market-wide stress and listing days.

The solver on real numbers: Yes Bank, 27 Sep 2019

	Quantity	Value	Where it comes from
Observed	Share price	Rs 48.75	NSE bhavcopy, 27 Sep 2019
Observed	Shares outstanding	231.5 cr	equity capital 463 / face value 2
Observed	Equity value E	Rs 11,286 cr	price x shares
Observed	Equity volatility sE	90.0%	250-day returns, split-adjusted
Observed	Default point D	Rs 335,982 cr	borrowings, FY2019 accounts
Solved	**Asset value V**	**Rs 324,420 cr**	**fixed-point iteration**
Solved	**Asset volatility sV**	**4.03%**	**fixed-point iteration**
Output	**Distance to default**	**0.72**	**standard deviations above debt**
Output	**Implied 1-year PD**	**23.4%**	**N(-DD)**

TABLE VII. Merton inputs and solution for a single date.

- Equity is worth only **3.4%** of the debt it stands behind, and is roughly twenty times as volatile as the assets — the option is far out of the money.
- The accounts at this date still scored 69. The market was pricing close to a **one-in-four** chance of default, five months before the moratorium.

Alert rule: level or deterioration

```
alert  =  FDI >= 80
       OR ( FDI >= 60  AND  FDI - median(own last 3 years) >= +15 )
```

- A level-only trigger fails on the case the exercise exists for. Yes Bank's FY2019 accounts scored **69** — below any sensible threshold, but **29 points above its own three-year median of 39.9**.
- The deterioration leg fires **8 months before** the RBI moratorium, at a cost of roughly three additional material names per year on the watchlist.
- A high level identifies *who* is weak; a sharp move identifies *when* something changed. An early-warning system needs both.

Case study: Yes Bank Ltd., FY2013-FY2020

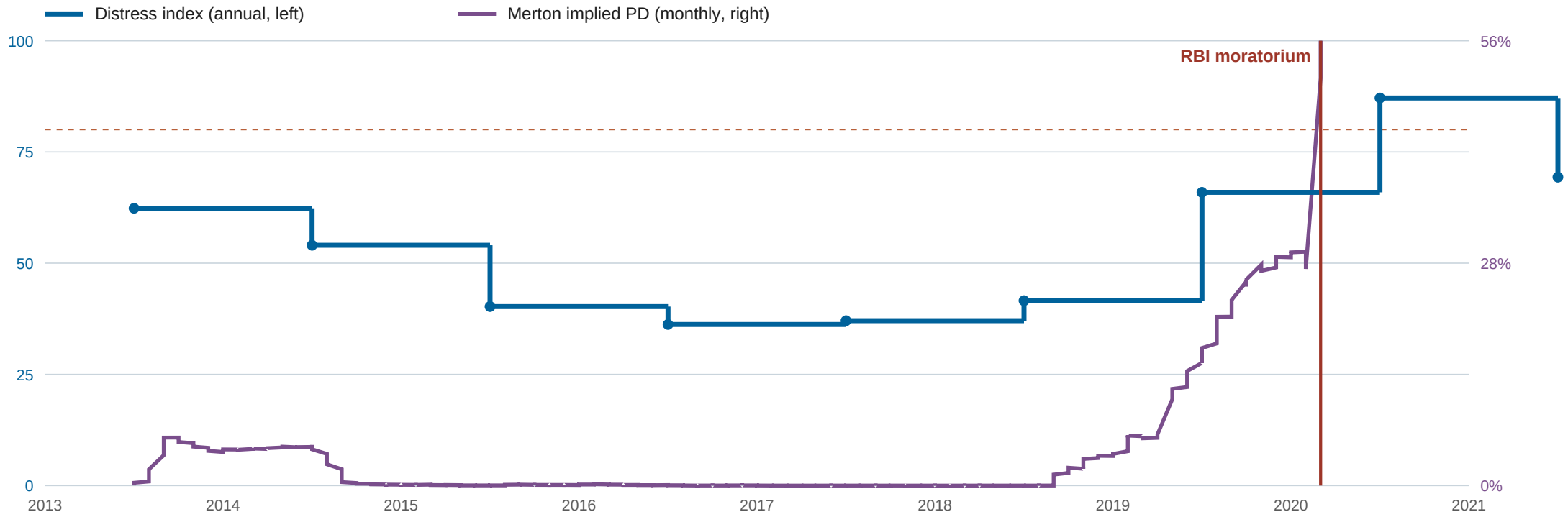

Fig. 1. Accounting distress index (left axis) and Merton implied one-year PD (right axis) on a common time axis.

The accounts reach 69 only on the FY2019 filings. The market had been pricing above 10% from mid-2019 and roughly one-in-three by January 2020 — the clearest evidence for running both signals.

Lead time across 18 labelled distress events

Institution	Event	Date	Lead (months)
Indian Overseas Bank	PCA	Oct 2015	39
Dhanlaxmi Bank	PCA	Nov 2015	40
United Bank of India ●	PCA	Jan 2017	42
Dena Bank ●	PCA	May 2017	10
IDBI Bank	PCA	May 2017	missed
Oriental Bank of Commerce ●	PCA	Oct 2017	3
Bank of India	PCA	Dec 2017	17
Allahabad Bank ●	PCA	Jan 2018	30
Reliance Home Finance	default	Sep 2019	2
Lakshmi Vilas Bank ●	PCA	Sep 2019	75
Yes Bank	moratorium	Mar 2020	8
SREI Infrastructure ●	resolution	Oct 2021	3
Reliance Capital	resolution	Nov 2021	41

TABLE VIII. Selected events (13 of 18 shown); a dot marks institutions recovered from source.

17 / 18
alerted before the event

14 / 18
within the 24-month window

52-94%
95% CI on recall

76
highest control reading

Does it generalise, or is it chance?

- **Unseen-event test.** Components re-selected on pre-2018 events only, methodology frozen, then run on seven later events: the same six components survive, and all **7 of 7** held-out events are caught.
- **Permutation test.** The null preserves event dates and sector, permuting only *which* institution failed, over 4,000 draws.

Statistic	Observed	Null mean	p-value
Events alerted (all 18)	17	7.20	< 0.00025
Alert precision	2.55%	1.02%	< 0.00025
Held-out events only (7)	**7**	**2.18**	**0.0003**

TABLE IX. Permutation test against a sector-preserving null.

Chance is ruled out as an explanation of *which* institutions are flagged. This does not establish performance on future events, and does not repair the four episodes still missing from the panel.

Accuracy is the wrong test for an alarm

Rule	Caught	AUC	Persistence	Always-on	Failed
Distress index (level or deterioration)	14/18	0.900	27%	2	0
Borrowings / equity, top decile	15/18	**0.976**	62%	**22**	0
Equity / assets, worst decile	17/18	0.932	60%	19	0
Merton DD, worst decile	4/18	—	23%	**0**	0

TABLE X. Alert utility. 'Always-on' counts institutions flagged in every one of 8+ observed years; 'Failed' counts how many of those actually failed.

- A single leverage ratio attains the **highest AUC** and is the **worst instrument**: it permanently flags HDFC Bank, ICICI, Axis, Canara and Bank of Baroda, and **none of the institutions it permanently flags ever failed**.
- It never changes its mind, so it carries no timing. **AUC is the wrong headline metric on a clustered event set**; lead time, persistence and watchlist size are what matter.

The same method, 547 US bank failures

Eighteen events cannot settle whether a method works. The FDIC publishes every US bank failure since 1934 and quarterly call reports for every insured institution, joined on one identifier. **Nothing was refitted** — same components, weights, anchoring, alert rule and horizon; only the field names change.

	India	United States
Panel	6,223 company-years	**416,285 bank-quarters**
Institutions	563	**10,341**
Period	FY2006-FY2023, annual	2003-2015, quarterly
Evaluable events	18	**547**
Pre-event observations	36	**4,309**
Survivorship bias	7 institutions recovered by hand	**none - failed banks remain**

TABLE XI. Scale of the external test.

Result on US data	Value
Failures alerted within 24 months	**535 of 547 (98%)**
Median lead time	**19.6 months** (quartiles 13.1 / 22.5)
Alert precision	7.32% against a 1.04% base rate - **7.1x lift**
Watchlist	829 of 8,005 banks per quarter (10%)
Permutation test	observed 535, null mean 141.9, **p < 0.001**

TABLE XII. Frozen methodology, unchanged, on 547 failures.

Two findings that change earlier conclusions

1. The AUC trap is settled. On Indian data a single leverage ratio beat the composite (0.976 against 0.912), defended as an artefact of an event set clustered on thin-capital public-sector banks. That defence is testable where no such clustering exists.

Signal	AUC	Caught	Persistence	Always-on
Distress index composite	**0.861**	**535 / 547**	**19%**	**2**
Leverage alone	0.828	511 / 547	27%	27

TABLE XIII. On 547 events the composite wins on every measure.

- The Indian ranking was the artefact it was argued to be. Leverage alone remains the worse alarm: **27** institutions permanently flagged against **2**.
- **2. The Indian weakness is a missing input, not a flawed structure.** Call reports carry a non-performing-assets ratio for every bank-quarter; as a standalone signal it scores **AUC 0.887 at 100% coverage** — stronger than any component in the model. The Indian archive has it only from 2020, for a tenth of the panel.
- **What this does not establish.** US and Indian banking differ in regulation, funding and resolution, and the US window is dominated by one crisis. This shows the method is not an artefact of the Indian sample; it does not show the Indian results are precise.

Signals that were tested and rejected

Candidate signal	Coverage	AUC	Added events	Watchlist cost
Market: drawdown from peak	20%	0.777	0	+17%
Market: abnormal 12-month return	17%	0.769	0	+11%
Funding: loan book shrinking	61%	0.736	+1	+57%
Funding: thin liquid buffer	100%	0.390	0	+74%
Quality: profit minus operating cash	98%	**0.321**	0	+85%

TABLE XIV. Candidate extensions. None was adopted.

- **Earnings-quality measures are inverted for lenders.** A lender's operating cash flow is dominated by loan-book movement, so a shrinking book releases cash: distressed lenders show cash *above* profit. The corporate-finance intuition does not transfer.
- Market signals discriminate well but flag institutions the accounts already condemn — they add *timing*, which the Merton layer already supplies, not new names.

What the evidence does not support

- **Eighteen events is a small sample**, and 13 are Prompt Corrective Action placements. The failure mode of greatest practical interest — an NBFC that stops paying — rests on five observations.
- **The index is a rank, not a probability.** A score of 88 means worse than 88% of peers, not an 88% chance of default. The Merton PD is a model output under strong assumptions, not a calibrated forecast.
- **Four distress episodes remain unevaluable** (DHFL, IL&FS, PMC Bank, Altico), and all four are failures, so measured recall remains optimistic.
- **No notion of sovereign or parent support**, and group or related-party exposure — the mechanism that actually brought down IL&FS and Reliance Capital — is invisible in a standalone balance sheet.
- **Component selection used the same events it was tested on**; the held-out test mitigates but does not eliminate this.

- Credit-risk techniques normally restricted to commercial providers can be reproduced from entirely public data, across **563 institutions** rather than a handful.
- The system alerted before **17 of 18** real distress events and within the actionable window for **14**, while healthy controls never exceeded 76 of 100.
- A sector-preserving permutation test rules out chance on held-out events (**$p = 0.0003$**), and the frozen method transfers to **547 US bank failures**, catching 535 with a median lead of 19.6 months (**$p < 0.001$**).
- The two analyses are complementary by construction: the accounts fail where losses are unrecognised, and precisely there the market moves first.
- **The most transferable finding is methodological** — on a clustered, rare-event problem, AUC rewards instruments that are useless as alarms. Persistence and watchlist size decide whether a model can actually be used.

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